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The leaching of salts from saline heavy clay soils: factors affecting the leaching process

T.W. Tanton, D.W. Rycroft & F.M. Wilkinson¹

Abstract. Cracks in dry saline montmorillonitic clay allow the soil to wet rapidly when flooded with negligible redistribution of salts. Once closed the only effective pathways remaining for the movement of leaching water are old root channels and faunal burrows. However, their effectiveness in conducting water and for leaching is severely restricted because of the lack of horizontal connections between them. Restructuring of clay can introduce sufficient permeability to a depth of about 0.7 to 0.8 m to allow salt to be leached provided that the soil's initial moisture content is sufficiently large to prevent disintegration upon wetting (about 24%) and that the clay is not allowed to become unsaturated during the leaching.

IN HOT dry countries the irrigation of cracking clay soil has resulted in the accumulation of salt and the abandonment of large areas of potentially fertile land. Conventional pipe drainage can control salinity, but because of the very close spacings required it is uneconomic, and if the land is to be kept in production a cheaper system must be developed.

Clay soils are characterized by having two distinct populations of different sized pores. One consists of the micropores within the clay mass, and the other comprises macropores formed by roots, fauna and by swelling and shrinking as the soil wets and dries. Significant drainage occurs only in the macropores (Bouma *et al.*, 1977; Bouma & Wosten, 1979). Hence, salt contained within the micropore water may be considered as immobile while that in the macropores can be considered as a mobile fraction. When dry clays are irrigated by flooding water spills down the cracks previously formed by drying and is absorbed rapidly by the dry soil. The infiltration rate decreases sharply as the soil swells and the cracks close (Germann & Bevan, 1981; McIntyre *et al.*, 1982) leaving only a few interconnected channels which remain as pathways for leaching.

There are two processes by which salt can move in clay, namely:

- (a) diffusion resulting from a concentration gradient, and
- (b) mass transfer with the seepage.

Diffusion is much too slow to be effective in transporting salt over long distances within the clay. Consequently large quantities of salt can be transported only by mass movement of water. Therefore, for leaching to be effective it is essential to maximize the mobile fraction of water. Diffusion does, however, play a vital role in transporting salt from the immobile water to the mobile water.

In this paper we evaluate the role of the natural macropores in leaching; concluding that the macropore structure must be improved if saline clays are to be reclaimed. In a second paper (Tanton *et al.*, 1988) we evaluate artificial restructuring of the soil as a means of improving the soils hydraulic properties and investigate the following questions:

- (i) To what extent will artificially induced structure be affected by the collapse of the soil which is known to occur when dry soil is rapidly wetted?
- (ii) To what depth will a restructured soil remain permeable?

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- (iii) Will the induced structure be retained for sufficient time to allow leaching to take place?
- (iv) How much salt will be leached out of a restructured soil?

MATERIALS AND METHODS

The soils

The movement of water and salt was studied using flood irrigation in a saline ($EC_e 10 \text{ dS m}^{-1}$) montmorillonitic clay soil on the Koy Hizmetleri Arastirma Enstitusu research farm within the Lower Seyhan Irrigation Project near Tarsus in southern Turkey. The field selected was drained by pipe drains 1.7 m deep and 40 m apart, though the drains had failed to control the salinity. The soil had a clay content exceeding 60% and was cracked at the surface into the characteristic hexagonal pattern associated with swelling clays. The cracks were up to 100 mm wide at the surface narrowing with depth until they terminated some 800 mm below ground level. The upper 150 mm of the soil was loose, dry and granular, in contrast to the massive sub-angular blocky structure of the subsoil. At 500 to 700 mm depth there was a dense band of small gypsum crystals.

Laboratory studies were conducted on soil consisting mainly of montmorillonitic clay (> 51% clay) of the Denchworth soil series. It was collected from a depth of approximately 1.0 m from a brickworks at Selbourne in Hampshire, England. In some experiments the soil was made saline by prolonged immersion in a salt solution of electrical conductivity, EC_e , of 8.4 dS m^{-1} , a sodium absorption ratio (SAR) of 3.58 (i.e. $5.0 \text{ g CaCl}_2 \cdot 6\text{H}_2\text{O}$; $2.52 \text{ g MgCl}_2 \cdot 6\text{H}_2\text{O}$; and 1.32 g NaCl per litre) until its cation composition became uniform.

Experimentation

The movement of water into dry cracked clay soil was studied in a field in Turkey by forming earthen bunds to enclose squares of area 6.25 m^2 . Six hundred litres of water, with a salinity of $EC_e 2.3 \text{ dS m}^{-1}$ and containing 0.5% by weight of methylene blue dye were supplied to the bunded areas at the rate of 20 l min^{-1} . Typically an area was left for 24 hours to allow the water to move into the soil, after which a trench was excavated to a depth of 800 mm across the middle of the plot and the faces of the trench examined for the presence of dye. Thereafter the soil was cut into horizontal slices 50 mm thick to allow the dye patterns to be studied in detail. Individual peds were also split apart to determine whether dye had entered them.

The rate of closure of cracks at the soil surface was studied in four similar plots by identifying three major fissures near the centre of each plot and by installing pegs into the ground adjacent to both sides of the cracks. The areas were then rapidly flooded, and the distances between the pegs were measured as the soil swelled until there was no further movement.

Water movement in saturated soil was studied in small square plots of area 6.25 m^2 which had been previously flooded for 48 hours. This initial wetting phase used some 2000 litres of water/plot. Methylene blue dye was then introduced into the ponded water remaining and flooding was maintained for a further 6 hours. However, all the dye became adsorbed by the upper 200 mm of topsoil. This layer was then carefully removed from one plot and the area reflooded. A 400-mm diameter infiltration ring was then placed in the middle of the flooded area and was filled with the dye solution. After 24 hours the ring was removed and the soil allowed to drain. The pattern of dye movement was then studied by excavating horizontal slices at progressively increasing depths below the surface.

The effect of the initial moisture content on the stability of clays under flood irrigation was studied using soil samples taken from below the root zone at a depth of about 1 m. The clay samples used were:

- (1) montmorillonitic clay from Selbourne in Hampshire, England (51% clay, 32% silt, and sand);
- (2) the same Selbourne clay which had been salinized by placing in a salt solution of SAR 3.58 and an EC_e of 8.4 dS m^{-1} for 35 days;
- (3) saline sodic montmorillonitic clay from Hoo Marsh in Kent, England. This soil has a large clay content (> 70%), an EC_e of 6.6 dS m^{-1} and an exchangeable sodium percentage (ESP) of 8. The soil readily disperses when it comes into contact with rainwater.

Cubes of volume 9 cm^3 were prepared, and their moisture contents adjusted to between 3 and 29% by air drying. The samples were dried in air until they reached a predetermined weight, when they were placed in sealed containers to allow the moisture to equilibrate throughout the samples. Aggregate stability was then measured by wet sieving (Yoder, 1936) through a series of sieves with 20-, 14-, 10-, 5- and 0.6-mm opening sizes.

In all cases 15 cubes were immersed in water ($EC_e = 9.4 \text{ dS m}^{-1}$) for 3 hours before being sieved to allow the clay to take in water. The wetted soils were then transferred to the sieves where they were gently agitated and continuously washed with a stream of tapwater until a stable equilibrium was established. The measured weight of clay retained on each sieve was then used to calculate a stability character termed the mean weight diameter (MWD). This is the summation of the product of the percentage weight retained on each sieve and the mean size interval between sieves. A large value of MWD indicates a stable soil; a small value corresponds to an unstable soil.

The stability of restructured soils under flooding was studied in cylindrical columns (0.14 diameter) packed to a depth of 750 mm with 1-cm^3 volume cubes of Selbourne non-saline montmorillonitic clay soil (moisture content 34% w/w). Manometers installed at intervals down the column enabled the head losses to be measured during downward flow and thus the hydraulic conductivities to be estimated at different levels until consolidation ceased. The

columns were flooded, and a small downward flow established. After consolidation had ceased the surface of the clay was loaded in successive stages to simulate the effect of increasing overburden weight, and the decline in hydraulic conductivity was determined at each loading.

The leaching of restructured saline clays was also studied in similar columns, but this time the clay had been fully salinized by immersion in water with an EC of 8.4 dS m^{-1} for one month. A downward discharge of approximately 1.25 ls^{-1} , corresponding to a downward flow rate of 8.7 m day^{-1} (using water of salinity $\text{EC} = 0.48 \text{ dS m}^{-1}$) was then imposed for a period of 15.5 hours. During this time the salinity of the effluent was measured.

The above procedure was repeated using larger cubes, each of volume 64 cm^3 for a period of 77 hours. In this case the downward flow rate was restricted to 0.62 m day^{-1} to simulate conditions likely to be encountered in practice.

RESULTS

Water movement in dry soil

When the soil was flooded water poured down the cracks. It entered the mass of the clay through the micropores and left a coating of dye adsorbed on the surface of the peds. Dye extended into the soil down to a depth of 0.8 m. Horizontal cracks within the clay peds also became coated with dye. The peds had a well-defined and apparently stable structure which could be readily identified in both the wet and dry state. However, upon wetting the peds disintegrated into smaller peds some 25 mm to 100 mm across which were themselves coated with dye. This secondary structure was not visible in the dry peds and is thought to have developed as the dry clay absorbed water. The development of these smaller peds was studied in more detail by placing a dry clay ped without visible cracks in a solution of dye. With no constraints to prevent swelling the ped fell apart as it absorbed water, ultimately reducing the single ped to many smaller ones.

The surfaces of the main peds were polished by having slid past one another, but there was no evidence of such movement between the subpeds. Dye was also deposited in

faunal burrows and root channels, emphasizing their importance for the movement of water within the soil.

Figure 1 illustrates the rapid closure of cracks that occurs under flood irrigation. The cracks became virtually closed within one hour.

Water movement into saturated clay soil in the field

In saturated soil patterns formed in the soil by the dye were recorded by mapping the dye's distribution at increasing depths below the surface.

The results indicated that most of the dye moved into the subsoil down old root channels. At 400 mm depth about 3% of the cross-sectional area was stained, and much of the dye occurred in root channels of about 2 mm diameter, typically spaced about 30 mm apart. At 600 mm depth dye was confined to smaller root channels (about 1 mm diameter) occupying less than 0.1% of the cross-sectional area and spaced at about 100 mm intervals apart. The dye seemed not to have moved horizontally. There was no dye below 800 mm.

Stability of restructured soil

Figure 2 shows the relationship between aggregate stability upon wetting and the initial moisture content of the soil. All soils remained fairly stable when their moisture content exceeded about 24%, while non-saline, clay soil remained stable until the moisture content became less than 16%. As might be expected saline soils were less stable than the non-saline with the saline-sodic soil being the least stable.

Hydraulic conductivity of restructured clay

In this experiment the effect of increasing weight on the hydraulic conductivity of restructured clay soil was studied under flooding. During the first test with no externally applied load the unloaded uppermost layer retained its integrity with a hydraulic conductivity exceeding 3000 m d^{-1} . The hydraulic conductivity of the lowest most heavily loaded layer (load approximately 2500 N m^{-2}) decreased to around 1000 m d^{-1} . The addition of an external load of 2590 N m^{-2} caused the hydraulic conductivity of the lowest layer to decrease to around 10 m d^{-1} . A further increase in the external load to 6600 N m^{-2} reduced the overall hydraulic conductivity of the column to 2 m d^{-1} , the final loading being roughly equivalent to that in an undisturbed soil at 0.7 to 0.8 m. At this stage flow was confined to just a few pathways instead of the rather uniform bathing of the cubes observed initially. This bypassing can be expected to reduce considerably the efficiency of leaching.

These experiments were limited to uniform aggregate size. In practice mechanized restructuring may be expected to produce a much more diverse soil structure varying from the mixed tilth of cultivated layers to the system of large fissures formed by deep subsoiling. Our results suggest that enhanced permeability may be expected to remain in the

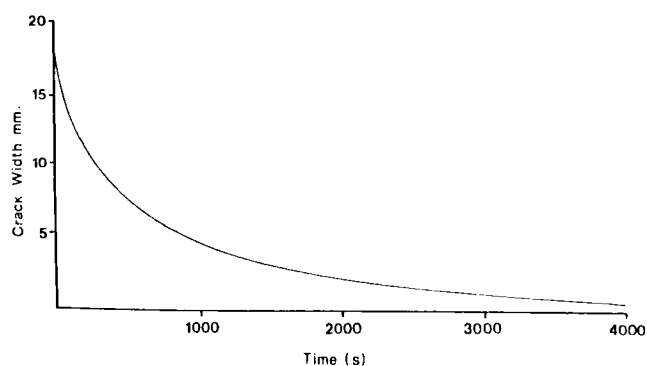


Fig. 1. Closure of cracks upon flood irrigation.

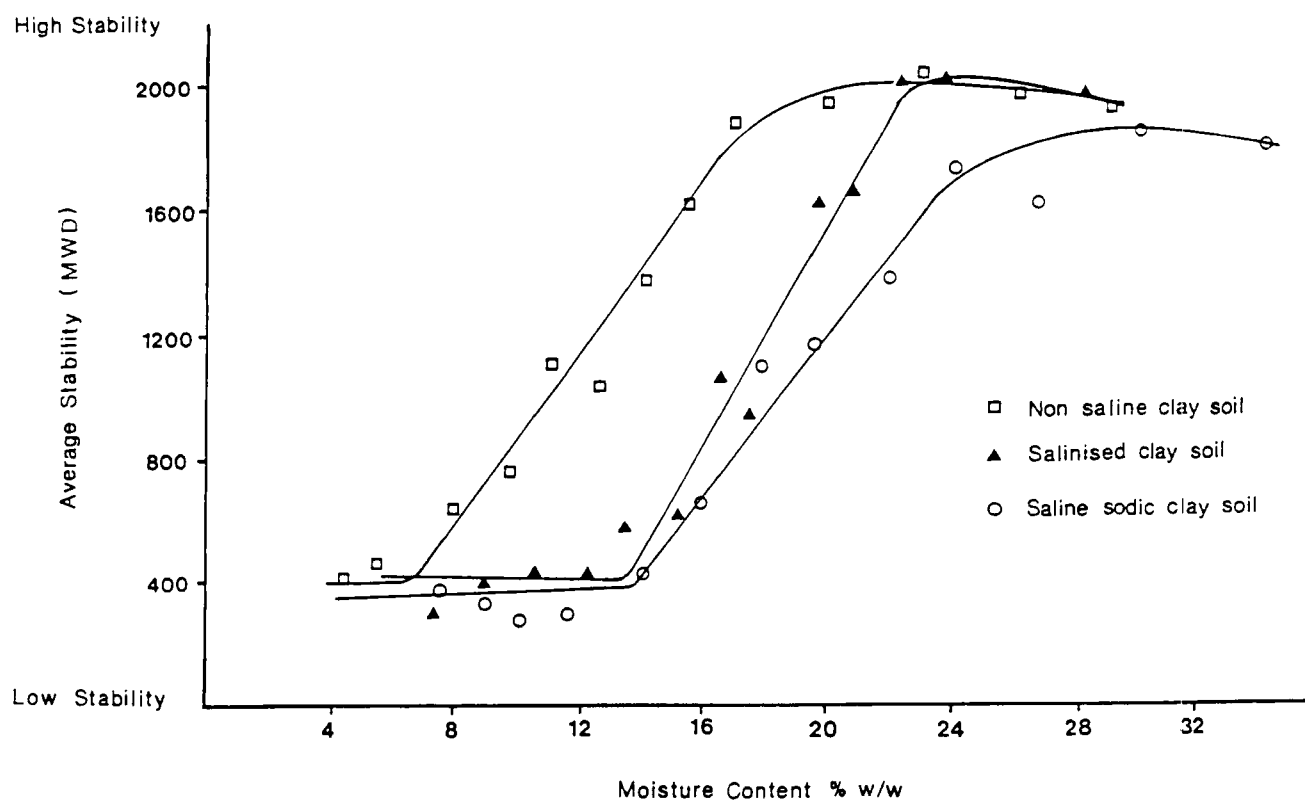


Fig. 2. Relationships between moisture content of clays and their aggregate stability when wetted.

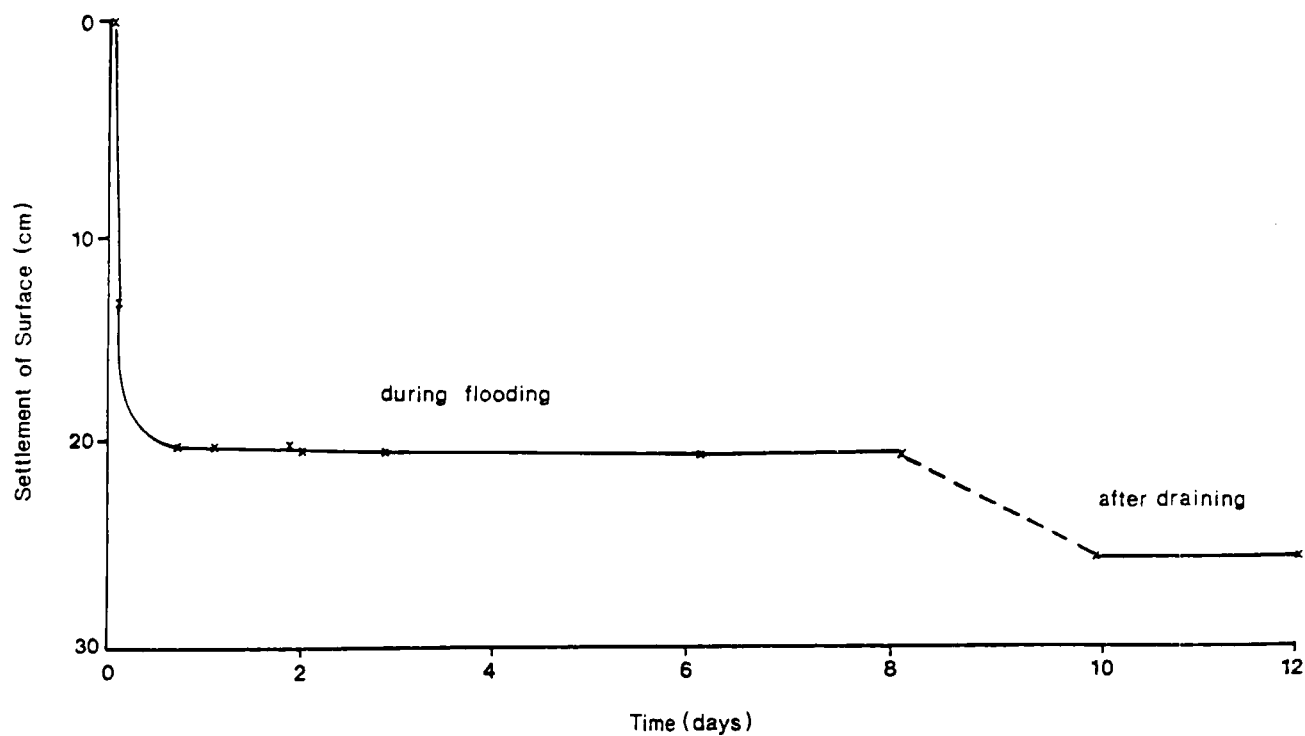


Fig. 3. Settlement of the surface of the soil in a large lysimeter during flooding.

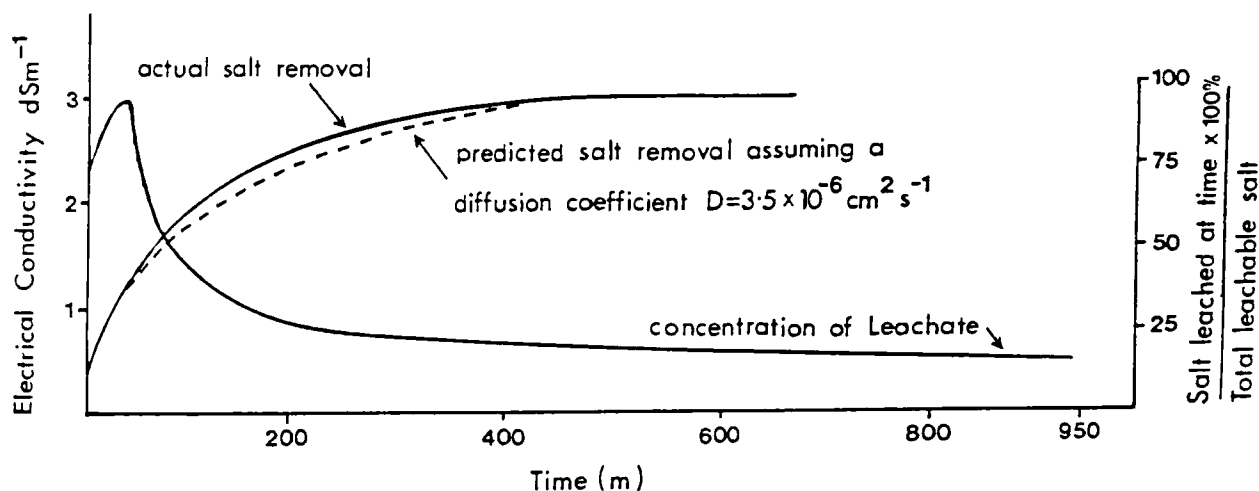


Fig. 4. Electrical conductivity and fractional salt removal during leaching.

upper layers of the soil whilst further down the weight of the soil combined with swelling pressure would reduce hydraulic conductivity fairly rapidly. Restructuring to a depth of about 0.8 m might be expected to improve the hydraulic conductivity enough for drainage to occur through what would otherwise be impermeable soil. The importance of keeping a restructured soil flooded to maintain the macrostructure was investigated on a saline Selbourne clay soil ($EC_e = 18.7 \text{ dSm}^{-1}$) contained in a large cylindrical lysimeter (1.25 m deep and 0.8 m in diameter) which had been dug out by spade and repacked. The lysimeter was flooded and allowed to drain from its base. The settlement occurring during the first 12 days of flooding is shown in Fig. 3. This shows that the soil settled rapidly initially and then remained fairly stable until the water table was drawn down, when a further 5% of settlement occurred.

Leaching of salts

The measured salinity of the drainage water and the fraction of salt leached from restructured soil contained in a 0.75-m deep column is shown in Fig. 4. Also shown is the predicted rate of leaching from spheres of 10 mm diameter into an infinite body of water. The prediction is based upon mathematical solutions of Fick's differential equation of diffusion for leaching from (saline) spheres into an infinite volume of water having constant and uniform salinity (Crank, 1975). During this test the hydraulic conductivity of the soil remained large ($> 10^3 \text{ m d}^{-1}$).

The tests involving impeded drainage from a column containing 40-mm sized saline clay aggregates yielded the expected pattern of increase and decrease in the salinity of the drainage water (Fig. 5) with the rate of removal of total soluble salts being less than the predicted rates for diffusion into an infinite volume of water (Fig. 6).

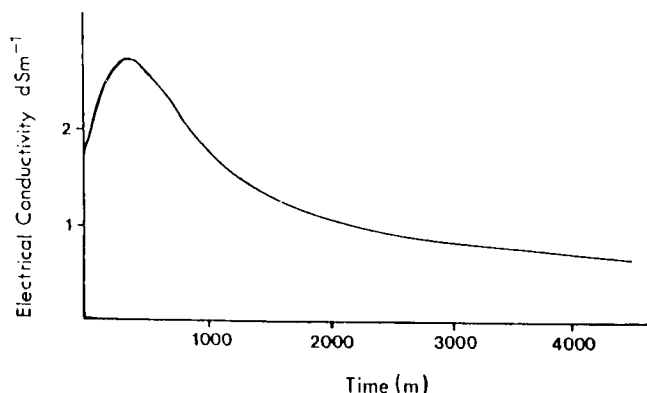


Fig. 5. Salt removal from clods of 40 mm diameter with restricted downward flow.

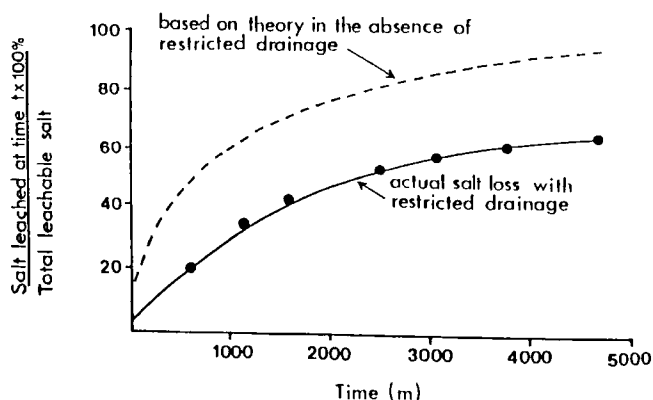


Fig. 6. Fractional salt removal from 40-mm diameter clods in a column with restricted drainage.

DISCUSSION

When a dry and cracked montmorillonitic clay soil was flooded water rapidly distributed itself throughout the soil

profile via the large cracks between the peds. The dry unconfined soil within the plough layer disintegrated into smaller aggregates and developed an intricate network of macropores that would allow rapid leaching of salt. Below the plough layer irrigation water flowed through large interped cracks and entered small cracks within peds which evidently developed on wetting. The water was then absorbed by the micropores within the clay matrix. This behaviour caused water to be fairly uniformly supplied to all parts of the soil and ensured rapid wetting and swelling. Since most flow bypasses the main mass of the soil it is to be expected that there will be little redistribution of salt during the wetting phase. Our study of the patterns of salt distribution both before and after flooding confirmed this. It also revealed that some salt moved downwards from within the plough layer, but there was no redistribution of the salt contained within the sub-soil peds. In addition, the rapid closure of the soil cracks prevented them from being of any value for leaching of the soil.

After the initial wetting water entering the soil moved freely through the intensive macropore system within the plough layer. Below this layer the structural cracks had closed so that water movement was confined to biopores, a phenomena also noted by Bouma *et al.* (1977). If the channels were to be freely drained from their base then their distribution would be sufficient to control salinity. This is demonstrated in Fig. 7 where the rate of salt removal for the above condition is predicted for different channel spacings (Barron, 1948). The model assumes that salt moves by diffusion from the saline clay mass to the freely draining channels where the rapid removal of water ensures a small salt concentration within the channels, thus maximizing the gradient for salt movement by diffusion. Unfortunately, as we found in Turkey, such channels terminate at the limit of rooting and so they cannot conduct water away. The remaining alternative for leaching would be in the horizontal

flow to a drain, but we have found that there are far too few horizontally interconnected pores to provide adequate drainage and leaching. In view of these difficulties it is felt that the only practical solution for leaching such soils is to create macropores artificially to enhance horizontal hydraulic conductivity and water movement to drains.

If an artificial structure is to be introduced it is important to minimize the consolidation occurring when the soil is flooded. In dry climates the soil moisture content of uncropped soil frequently falls to less than 20% (w/w) with moisture contents around 10% being common in the plough layer. Our results indicate that at these small moisture contents the soil can be expected to slake, leading to a considerable loss of permeability. The moisture content should where possible be maintained at 24% or more to avoid this happening.

Laboratory measurements of the hydraulic conductivity of restructured soil indicate that clay with naturally poor permeability ($< 0.05 \text{ m d}^{-1}$) retained sufficient induced permeability to provide effective drainage and leaching. In this case rapid settlement of the restructured soil occurred when it was wetted. Soil at the surface retained a very large hydraulic conductivity ($> 3000 \text{ m d}^{-1}$) while soil subjected to a loading equivalent to 0.7 to 0.8 m of undisturbed soil retained a hydraulic conductivity of 10 m day^{-1} .

The large hydraulic conductivity of the restructured clay enabled salt to be leached from the column rapidly and efficiently. When a large volume of water passed through the soil column the rate of salt removal was close to that predicted by diffusion theory using a diffusion coefficient of $3.5 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$, a figure previously determined experimentally (Tanton *et al.*, 1987). When the drainage was restricted to a value corresponding more closely to rates encountered in the field the rate of salt removal was much less than that predicted by diffusion theory. In this case and within natural clay soil in the field the mass flow of water limits leaching.

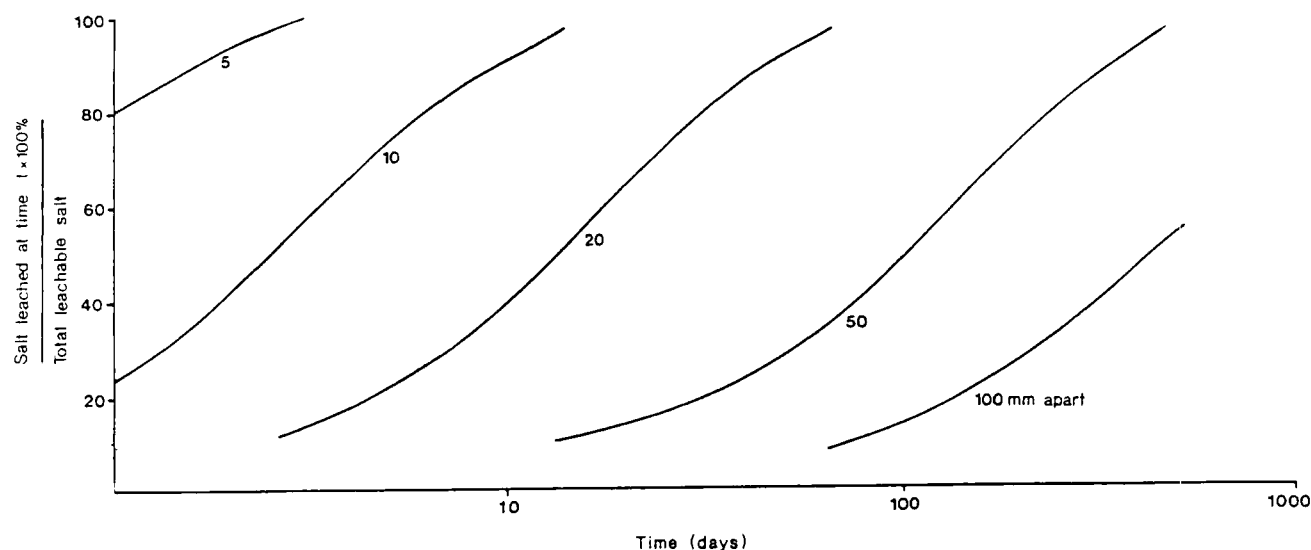


Fig. 7. Predicted rate of removal of salt from soil drained by vertical channels at various distances apart.

Overall it should be feasible to restructure a saline clay soil to a depth of 0.7 to 0.8 m and retain sufficient structure to allow enough lateral flow to occur to an installed drainage system and thus leach a saline clay soil.

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The leaching of salts from restructured saline clay soils

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Abstract. The leaching of salt from a restructured saline heavy montmorillonite clay soil was studied experimentally. Restructured clay was placed in a 25-m long $\times$ 0.75-m deep polythene-lined trench and leaching studied under flood irrigation. The hydraulic conductivity of the soil always exceeded 25 m d⁻¹ during the experiment, and 85% of the salt was leached within 16 days, most being leached in the first 8 days. The results indicate that restructuring clay could make the drainage of heavy clay soils economically feasible.

THE INHERENT fertility of heavy textured soil makes it very attractive for agriculture, and in the past 50 years much of such land has been irrigated. Unfortunately in dry climates the poor internal drainage of such soil has caused an accumulation of salts, with the result that there are now vast areas of saline clay soil needing reclamation. When the depth of clay extends beyond any reasonable drain depth the economics of reclamation are very unfavourable. Previous work indicates that if such soil is to be economically reclaimed by pipe drainage then it will be essential to improve the permeability of the soil by artificially restructuring it (Tanton *et al.*, 1987, 1988).

Soil restructuring in combination with mole drainage has been practised for many years. Unfortunately, although such a technique can improve drainage, it is of little value for leaching salts because it results in a rapid bypassing of water to the drain, providing little opportunity for leaching.

Tanton *et al.* (1988) indicated that a restructured soil can retain sufficient induced permeability to allow effective leaching to occur over fairly short times. The experiment

described in this paper was therefore designed to provide guidelines for the design of a field leaching system. Lateral seepage and leaching in a restructured soil were studied in a lined narrow trench, a procedure designed to remove the variables of downward and sideways movement which can confound the results of drainage trials.

MATERIALS AND METHODS

The site

The studies were conducted on a site within the 131 300 ha Seyhan Irrigation Project near Tarsus in southern Turkey. The soils in the area range from silts in the upper regions of the Seyhan plain to montmorillonitic clays near the Mediterranean coast.

The field selected had a large clay content (>60%), was saline (EC_e 6.14 dS m⁻¹) and non-sodic with large quantities of precipitated gypsum below 0.35 m. Pipe drainage at 1.75 m depth and at 40 m spacing had been installed some

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